

Waiting for GPT: Probing Interaction Patterns in High-Latency RAG Chatbots

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Abstract

As AI systems shift toward compute-intensive reasoning, traditional interface paradigms are being challenged. This work presents an exploratory case study investigating emerging interaction patterns for AI interfaces, using high-latency Retrieval-Augmented Generation (RAG) as a representative probe. We explore a subset of a broader design space focused on three critical dimensions of latency feedback, source transparency, and confidence communication. Through a pilot study, we probe how specific interaction patterns within this design space impact user experience. Findings suggest that in high-latency scenarios, users prioritize informative transparency over traditional metrics of perceived speed, which helps inform initial design considerations tailored to the emerging era of transparent, reasoning-heavy AI interfaces.

CCS Concepts

• **Human-centered computing** → **User interface design**; *Interaction design*; *Ubiquitous and mobile computing design and evaluation methods*.

Keywords

retrieval-augmented generation, chatbots, design patterns

ACM Reference Format:

Tobias Kothbauer, Kathrin Probst, and Andreas Stöckl. 2026. Waiting for GPT: Probing Interaction Patterns in High-Latency RAG Chatbots. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799266>

1 Introduction and Background

As human-computer interaction evolves, the shift from graphical to conversational user interfaces necessitates a new design approach focused on facilitating natural and trustworthy dialogue with intelligent agents [5]. Recent advancements in Large Language Models have led to increasingly sophisticated conversational agents, one of the most promising architectural patterns being Retrieval Augmented Generation (RAG) [8]. While this pattern mitigates common model issues such as factual inaccuracies and hallucinations by retrieving information from an external knowledge base

to ground generated responses [7], it introduces significant frontend challenges such as feedback latency, referencing of underlying knowledge sources, or indication of confidence.

Each of these frontend challenges requires specific design considerations. While system response time critically impacts user experience [10], research suggests that appropriately timed delays can reduce user frustration or make the interaction feel more human-like [6, 16]. Providing an explanation or showing a clear sense of progress during these waits can significantly improve perceptions of transparency and trust because justifying the delay makes the wait feel more tolerable and productive [18]. Beyond managing wait times, effective RAG interfaces must ensure the generated content itself is transparent and verifiable, which requires a human-centered roadmap where the interface explains its generative process by explicitly displaying its sources [9]. Furthermore, research emphasizes that showing exactly where information comes from and how it is grounded helps users feel more confident in the chatbot's responses and view them as more factual [4]. Similarly, communicating the system's certainty is a key design consideration. Beyond general human-computer interaction principles [11, 14], guidelines for Human-AI interaction emphasize the importance of making system capabilities and confidence clear to users [1, 13]. Empirical studies reinforce this, suggesting that displaying confidence scores in AI-assisted interfaces improves decision accuracy and strengthens user trust, particularly in environments where interpretability is key [12, 17]. Ultimately, addressing all three of these dimensions is crucial for building user trust that the agent will act reliably to help them achieve their goals.

This exploratory case study utilizes the RAG framework as a representative design probe to investigate interaction patterns that balance technical latency with the fundamental user need for transparency and trust. Specifically, we explore three core dimensions that define the temporal and transparent nature of the user experience in reasoning-heavy AI interfaces:

- *Latency Feedback*: How can the interface keep users engaged and informed during long response delays?
- *Source Transparency*: How can the interface link generated statements to their sources?
- *Confidence Communication*: How can the interface communicate certainty about its response?

By investigating these dimensions through a pilot evaluation, we aim to contribute an initial set of design considerations that help inform the development of more effective and user-centered AI interfaces that balance technical accuracy with the growing need for interpretability and transparency.



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799266>

2 Methodology

An effective RAG chatbot must not only provide accurate information but also present it in a way that is transparent, trustworthy, and engaging — especially when users are forced to wait. To investigate these aspects, we explored a variety of state-of-the-art UI patterns for the three design dimensions of latency feedback, source transparency, and confidence communication (see Figure 1), as these represent the primary friction points in modern information-seeking tasks. Our selection was conceptualized by drawing inspiration from established user interface design patterns [15] commonly used in contemporary commercial AI chatbots.

- For the latency feedback dimension, we aimed to contrast generic visual placeholders with informative, semantic feedback that explicitly visualizes the system’s reasoning process. To achieve this, we compared a pulsing *Skeleton* loader, a *Chain-of-Thought* visualization displaying internal processing steps, and a *Progress Bar* with percentage indicator.
- For the source transparency dimension, we explored the trade-off between uninterrupted reading experiences and immediate contextual transparency. To represent this, we chose to compare a *Tooltip* visualization appearing on hover, an *Accordion* variant expanding upon a click, and static *Cards* being displayed below the system response.
- For the confidence communication dimension, we wanted to compare simple numerical familiarity against multi-channel visual cues. Therefore, we decided to compare a plain *Label* displaying confidence in percentage, a *Badge* indicating confidence via color and text (low / medium / high), and a *Gauge* combining percentage, color, and symbolic representation.

We developed a functional prototype to serve as a representative design probe for evaluating these interaction patterns. The prototype was built as an interactive web application simulating a realistic chatbot interface. The backend was developed in Python using the LangChain library and the OpenAI GPT-4o model, while FAISS enabled efficient vector search and retrieval across a curated knowledge base containing administrative and technical text documents. To simulate a realistic condition of advanced reasoning models, we implemented a fixed 15-second delay for every response as typically experienced with high-latency RAG systems [3].

To compare these design dimensions, we conducted a pilot study, in which participants interacted with the prototype in a controlled session. A total of 13 participants (seven female, six male; aged 22-27 years, $Mdn = 24$) took part in the study, all of whom were frequent users of chatbots, such as ChatGPT. They were asked to perform specific exploratory tasks designed to probe system capabilities (e.g., retrieving specific information about university statutes, gathering details on broad topics such as JavaScript). For each challenge, participants completed a Likert scale questionnaire to rate usability, clarity, and trust before ranking the variants for each design dimension by preference. To prevent ordering effects, the presentation sequence of the UI variants was counterbalanced across participants. Each session lasted approximately 45 minutes and was concluded with a semi-structured interview to gather qualitative insights. The discussions were audio-recorded, transcribed verbatim, and then analyzed using reflexive thematic analysis [2] to identify recurring patterns.

3 Results

Participants’ preference ratings are summarized in Figure 2, further quantitative results are provided in the supplementary materials. We provide an integrated analysis of these rankings alongside the underlying qualitative findings in the following:

For latency feedback, the *Chain-of-Thought* indicator was the overwhelming favorite, ranked first by 9 of 13 participants because it made the wait feel transparent and productive. Participants noted that this visualization “showed clearly what the system was working on” (P3) and provided the “best confidence to know that the system is actively working” (P9), effectively reducing perceived delay duration. Conversely, the *Skeleton* loader was the least preferred and was described negatively by all 13 participants, who reported that they “did not see any progress” and consequently “felt stuck” (P13), while others noted that having “no further information” was “frustrating” (P1). A Friedman test confirmed significant differences in preference rankings ($\chi^2(2) = 12.15, p = .002$) with pairwise comparisons showing the *Chain-of-Thought* was significantly preferred over the *Skeleton* loader ($p < .05$). Overall, these findings suggest that for RAG pipelines experiencing high retrieval and synthesis latency, exposing internal processing steps as semantic feedback is a foundational pattern for maintaining user engagement.

For source transparency, the *Card* design was preferred by 8 of 13 participants, receiving the highest ratings for ease of use and visual clarity. Participants praised it for its efficiency because they could “see immediately the reference to the source” (P2), while the other designs introduced additional interaction costs. The *Tooltip* variant was perceived as frustrating because it required “precise hovering” (P5), the *Accordion* was criticized for being disruptive because it caused disruptive layout shifts when expanding the UI, which some participants mentioned made it “hard to focus” on their reading (P11). Statistical analysis showed a significant effect of the design variation on rankings ($\chi^2(2) = 12.15, p = .002$) as the *Card* design was significantly more preferred than the *Accordion* ($p < .05$). Overall, these findings indicate that zero-interaction transparency patterns are superior for maintaining cognitive flow in verifiable RAG environments. However, it is important to note that while the *Card* design is highly effective for low-density responses, scaling this UI pattern for complex RAG outputs containing dozens of references remains an open challenge.

For confidence communication, the *Gauge* and *Label* designs were strongly preferred over the *Badge* design, which 12 of 13 participants ranked last. Participants valued the clarity of a precise number as it combined the “arrow with the percent and the color” (P4), providing intuitive visual indicators. Those who preferred the *Label* design appreciated being able to make their “own interpretation” of the result (P10), validating our design rationale to display raw numerical scores rather than subjective text labels that might misalign with individual user thresholds. The *Badge* design was seen as ambiguous and raised accessibility concerns regarding color blindness (P6, P8). We found a significant difference in preference across scales ($\chi^2(2) = 16.62, p < .001$) where pairwise comparisons showed that both the *Gauge* and *Label* options significantly outperformed the color *Badge* ($p < .01$). Overall, these results suggest that explicit, accessible patterns for uncertainty communication are critical for establishing user agency in probabilistic RAG outputs.

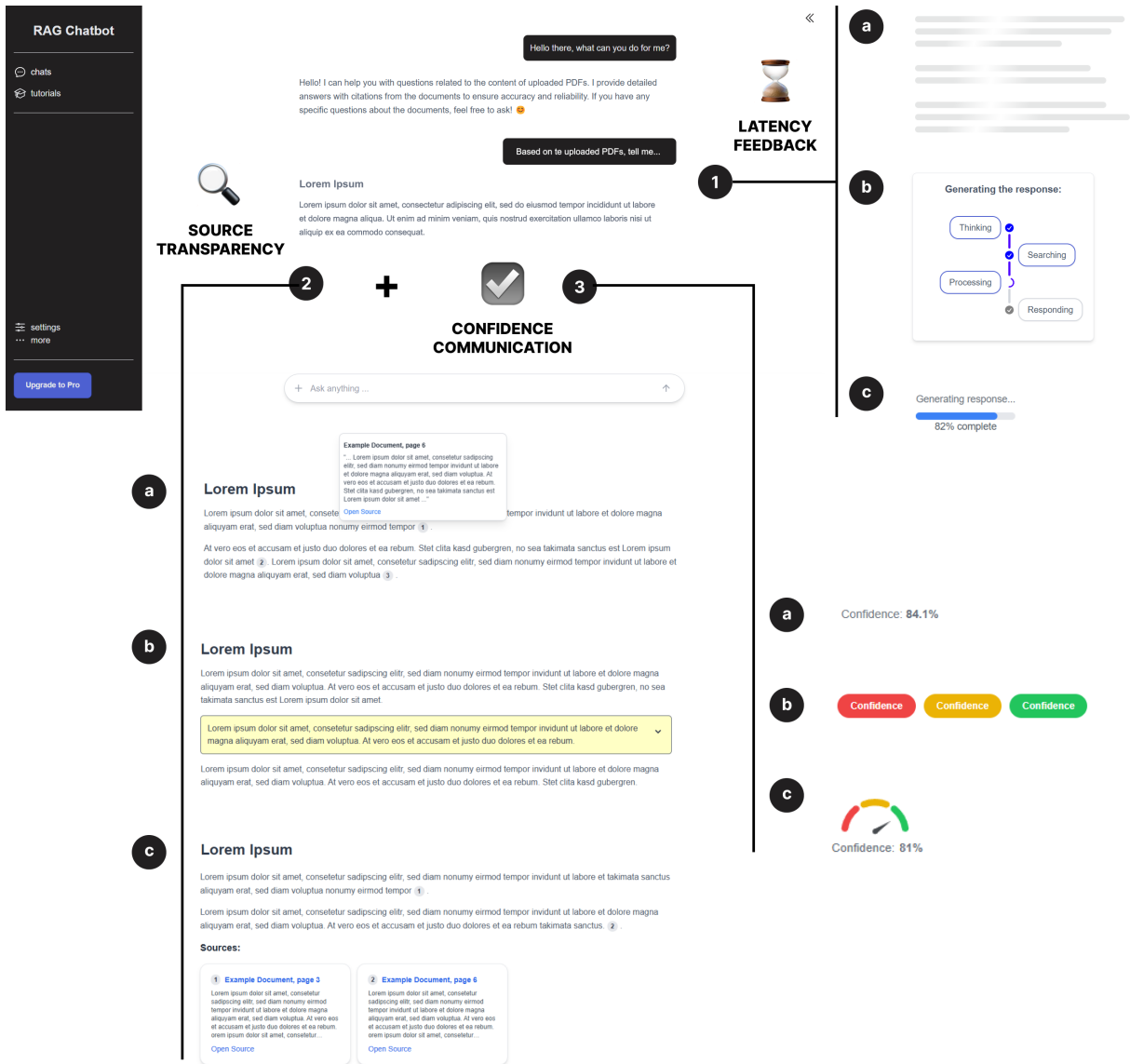


Figure 1: UI pattern variants implemented for (1) latency feedback: a) Skeleton, b) Chain-of-Thought, c) Progress Bar; for (2) source transparency: a) Tooltip, b) Accordion, c) Card; and for (3) confidence communication: a) Label, b) Badge, c) Gauge.

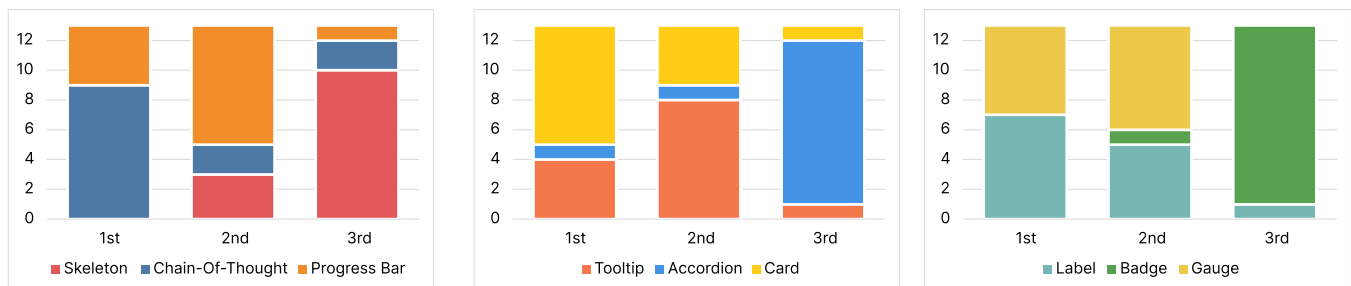


Figure 2: Preference ratings for the latency feedback (left), source transparency (middle) and confidence communication (right).

4 Conclusion & Future Work

The results of our pilot study converge on several key themes: users prioritize transparency, minimal cognitive and interactional load, and informative feedback. Interfaces that present information clearly and upfront, without requiring extra clicks or hovers, are perceived as more usable and trustworthy. During inevitable delays, feedback that explains *why* the user is waiting is vastly superior to a generic loading animation.

Based on these insights, we propose the following preliminary design implications, which may be considered as early-stage hypotheses meant to guide the design of high-latency RAG chatbots and spark further validation studies:

- *Expose semantic processing through active progress indicators.* To maintain user engagement during long retrieval and synthesis delays, interfaces should prioritize visualizing *what* the pipeline is doing over merely showing *that* it is working. In our probe, participants preferred a chain-of-thought visualization indicating the internal reasoning process, which made the wait feel more transparent and productive. Conversely, progress bars lacked the contextual clarity of step-by-step indicators, and non-semantic indicators like skeleton screens led to a perception of the system being stuck.
- *Prioritize zero-interaction provenance.* To support cognitive flow during fact verification, interfaces should present evidence in a clearly visible manner that allows users to directly access the grounding data. In our probe, participants preferred static cards that presented all source information upfront, while the dynamically displayed sources in the tooltip or accordion variants were found to increase interaction costs with additional hovers and clicks, or cause disruptive layout shifts upon expanding over the interface.
- *Support user-led interpretation through granular numerical scores.* To empower users to calibrate trust based on their specific task context, designers should present system certainty neutrally to allow users to define their own confidence thresholds. In our probe, participants preferred the textual percentage and the gauge visualization over the categorical color badges. This suggests that while raw numerical data is essential for nuanced judgment, using multi-channel visualizations that provide additional visual or symbolic cues could be used to improve clarity. Designers should thus prioritize textual accuracy while offering customization options that allow users to adjust thresholds for categorical feedback to ensure the interface remains adaptable to the specific demands of their domain.

While the preliminary nature of this study limits immediate generalizability, specifically due to the small and homogeneous participant pool and the use of fixed response delays, we think that the pilot insights derived from this probe highlight a need for new design patterns and heuristics in the next generation of more and more complex, reasoning-heavy AI interfaces.

Rather than being exhaustive, our findings serve as a catalyst for several major research trajectories. In an immediate next step, we aim to move beyond the initial UI patterns explored in this design probe to develop a more comprehensive and dynamic grammar of temporal design patterns. This work will involve validating our

preliminary heuristics with larger, more diverse participant pools to ensure the patterns remain effective across varying levels of user expertise. Second, we intend to investigate how these heuristics generalize beyond conversational RAG into other high-stakes AI environments with similar transparency requirements, such as AI-powered code editors or medical diagnostic tools where source verification and wait time remain critical friction points. Finally, by testing these interfaces in real-world scenarios, we seek to establish robust design standards that ensure as AI systems become more computationally complex, our interfaces remain clear, trustworthy, and fundamentally human-centered.

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